

ESED: Emotion-Specific Evidence Decomposition for Uncertainty-Aware Multimodal Emotion Recognition in Conversation

Zechang Xiong
Beijing Jiaotong University
Beijing, China
zcxiong@bjtu.edu.cn

Zhenyan Ji*
Beijing Jiaotong University
Beijing, China
zhyji@bjtu.edu.cn

Wenkang Kong
Beijing Jiaotong University
Beijing, China
wenkang.kong@bjtu.edu.cn

Jiuqian Dai
Beijing Jiaotong University
Beijing, China
daijiuqian@163.com

Shen Yin
Norwegian University of Science and
Technology
Trondheim, Norway
shen.yin@ntnu.no

Abstract

Multimodal emotion recognition in conversations is inherently challenging due to ambiguous cues, modality conflicts, and temporal dynamics, all of which contribute to complex and diverse uncertainty sources. While some recent methods incorporate uncertainty modeling, they often focus on overall prediction confidence, without explicitly distinguishing the different sources of uncertainty introduced by underlying factors. To address these challenges, we propose a novel Emotion-Specific Evidence Decomposition framework (ESED) that leverages evidential deep learning to explicitly model and disentangle multimodal emotional uncertainty. Rather than directly fusing features, ESED decomposes each modality's evidence into three interpretable components: (1) emotion-consistent evidence, capturing shared emotional cues across modalities; (2) emotion-specific evidence, highlighting the unique emotional role of each modality; and (3) dynamic evidence, modeling utterance-level temporal variations. These components are adaptively weighted based on emotional intensity, ambiguity, and dynamicity, quantified via prediction entropy, inter-modal divergence, and temporal variance. The final prediction is obtained through an adaptive fusion of these weighted components. Extensive experiments demonstrate that ESED outperforms the state-of-the-art methods on the MELD and IEMOCAP datasets, demonstrating the effectiveness of our proposed method.

CCS Concepts

• **Information systems** → **Multimedia information systems**; • **Computing methodologies** → *Discourse, dialogue and pragmatics*.

*Corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CIKM '25, Seoul, Republic of Korea.

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-2040-6/2025/11

<https://doi.org/10.1145/3746252.3761430>

Keywords

Emotion Recognition in Conversation, Evidential Deep Learning, Multimodal Fusion

ACM Reference Format:

Zechang Xiong, Zhenyan Ji, Wenkang Kong, Jiuqian Dai, and Shen Yin. 2025. ESED: Emotion-Specific Evidence Decomposition for Uncertainty-Aware Multimodal Emotion Recognition in Conversation. In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM '25)*, November 10–14, 2025, Seoul, Republic of Korea. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3746252.3761430>

1 Introduction

Multimodal emotion recognition in conversations (MERC) aims to infer speakers' emotional states by leveraging diverse information sources, typically including textual content, vocal cues, and visual expressions [9]. This task requires not only recognizing subtle and context-dependent emotional cues [15, 18], but also dynamically interpreting them across modalities and temporal sequences [34]. These inherent complexities make MERC a central yet challenging problem in conversational AI and human-centered computing.

Despite its potential, MERC remains fundamentally challenging due to the presence of multi-source uncertainty [1]. Building upon prior observations in MERC and drawing insights from conversational dynamics, we identify three principal forms of uncertainty that jointly define the recognition landscape. First, emotional expressions often exhibit inherent ambiguity, where a single utterance carries vague or context-dependent affective signals. As shown in Figure 1(a), the textual content alone may appear neutral or cooperative, yet when integrated with prosodic and visual cues, the underlying anger becomes evident. This highlights ambiguity as a core uncertainty source, necessitating multimodal disambiguation [34] for reliable interpretation [27]. Second, emotional signals across modalities may conflict, leading to modality-specific inconsistency. Figure 1(b) illustrates such a scenario: while the text conveys a positive sentiment ("I'm happy"), facial and vocal cues reflect suppressed or sarcastic emotions. This modality conflict introduces semantic tension between channels, forming a second key source of uncertainty that cannot be resolved through naive fusion [37]. Third, emotional states are intrinsically dynamic, evolving throughout a

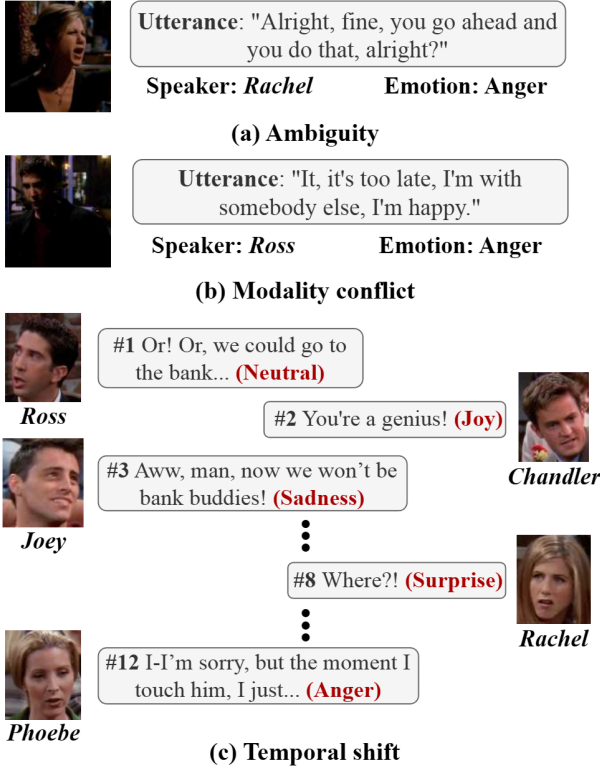


Figure 1: Illustration of three core challenges in MERC. (a) **Ambiguity:** The utterance contains emotionally ambiguous expressions that require context or disambiguation. (b) **Modality conflict:** Different modalities may convey inconsistent emotional cues. (c) **Temporal shift:** Emotions evolve across conversation turns.

conversation. As depicted in Figure 1(c), the speaker’s affect shifts across dialogue turns—from neutrality to joy, sadness, surprise, and anger. These temporal dynamics introduce uncertainty over time, requiring models to capture long-range dependencies and emotional trajectories beyond static representations. These three types of uncertainty construct a multi-dimensional uncertainty framework for MERC. Addressing them in a unified, interpretable manner is essential for developing models that are not only more accurate but also semantically aligned with human emotional reasoning.

In recent years, a variety of approaches have been proposed to tackle MERC’s inherent complexities [25, 29]. Early feature-fusion methods concatenate [3, 36] or jointly project multimodal embeddings, which can capture complementary cues but often overlook modality-specific nuances and temporal shifts. Attention-based [11] and transformer-driven models [25, 31] dynamically weigh modality contributions per utterance, improving alignment but still treating uncertainty as a uniform weight signal. Graph-based techniques model inter-utterance and inter-modal relations to address temporal dynamics and conflicts, yet they typically optimize for prediction accuracy without explicitly quantifying uncertainty. More recently, a few works have begun to integrate uncertainty modeling

[1] to estimate overall confidence in predictions. However, these approaches generally focus on a single, aggregated uncertainty score, failing to distinguish the underlying sources of ambiguity, modality conflict, and temporal variation, and thus offering limited robustness when heterogeneous or evolving signals dominate.

To address the above challenges, we propose a novel framework: Emotion-Specific Evidential Decomposition (ESED). Unlike conventional methods that rely on feature concatenation or undifferentiated fusion, ESED is built on the principle that emotional cues across modalities are inherently heterogeneous and that uncertainty should be modeled with source-specific semantic awareness. At the core of ESED is a structured evidence decomposition mechanism. Rather than treating multimodal signals as homogeneous indicators of affect, ESED explicitly decomposes the raw emotional evidence from each modality into three semantically grounded components, each corresponding to a distinct source of uncertainty in MERC: ambiguity, modality conflict, and temporal dynamics. **The emotion-consistent evidence** captures information that is aligned across multiple modalities, forming a consensus-based representation that reduces ambiguity. By emphasizing inter-modal agreement, this component mitigates uncertainty arising from vague or under-specified signals in individual modalities. **The emotion-specific evidence** isolates modality-unique emotional cues, enabling the model to identify and quantify cross-modal divergence. This component is particularly critical for addressing modality conflict, such as sarcasm in text, prosodic shifts in speech, or subtle facial expressions, which may provide contradictory or complementary signals. **The dynamic evidence** models the temporal evolution of emotional states across conversational turns. By explicitly encoding temporal dependencies, this component addresses uncertainty related to emotional transitions, allowing the model to capture both short-term fluctuations and long-term emotional trends. This evidence-oriented decomposition endows the model with an unprecedented ability to examine and reason over emotional cues at a finer granularity, enhancing both interpretability and adaptability in dynamic and multimodal settings. By embedding uncertainty modeling directly into the network architecture, rather than relying on post hoc fusion or heuristic weighting, ESED offers a principled, semantically interpretable, and empirically effective solution to multimodal emotion recognition in conversation.

Our main contributions are as follows:

- We identify and categorize three core sources of uncertainty in multimodal emotion recognition in conversations — *ambiguity*, *modality conflict*, and *temporal shift*—grounded in real-world dialogue scenarios.
- We propose a novel MERC framework, Emotion-Specific Evidence Decomposition (ESED), which explicitly disentangles emotion-related evidential components according to their uncertainty sources. This decomposition enables interpretable, fine-grained modeling of complex affective phenomena across modalities and time.
- Extensive experiments on benchmark datasets (MELD and IEMOCAP) demonstrate that ESED achieves state-of-the-art performance while offering better robustness under uncertainty compared to existing approaches.

2 Related Works

2.1 Multimodal Emotion Recognition in Conversation

Multimodal Emotion Recognition in Conversation [28] focuses on inferring speakers' emotional states by jointly analyzing heterogeneous signals from text, audio, and visual modalities [31]. This task poses challenges due to the inherently nuanced, context-dependent, and temporally dynamic nature of emotional expression [10, 29]. Recent research [6, 34] has primarily centered around sophisticated fusion strategies designed to integrate complementary information from each modality. State-of-the-art approaches leverage attention mechanisms [35], graph-based structures [16, 33], and transformer architectures [8, 17] to model inter-modal dependencies and conversational dynamics. However, many existing methods either assume equal importance across modalities [14] or employ late fusion strategies, thereby overlooking potential modality conflicts and inconsistencies—an issue particularly critical when modalities offer contradictory emotional cues [11]. Furthermore, temporal modeling remains underexplored. Most frameworks fail to recognize shifts in affective states over time. This has prompted recent efforts [20, 32] to develop more robust architectures that not only integrate multimodal signals but also disentangle their respective contributions based on emotional salience and contextual relevance. Emerging paradigms driven by LLMs are beginning to reshape the MERC landscape [5]. LLMs offer powerful capabilities as both encoders and reasoning engines, enabling deeper comprehension of emotional context, implicit intent, and social cues [7].

2.2 Uncertainty Modeling in Multimodal Learning

Uncertainty modeling has emerged as a crucial component in multimodal learning, particularly for improving model robustness under noisy, incomplete, or conflicting inputs. Evidential deep learning [30] introduced a principled framework based on subjective logic and Dirichlet distributions, enabling models to represent class probabilities and associated uncertainty in a unified formulation. Building on this foundation, a growing line of work has focused on resolving interview conflicts by integrating uncertainty into the fusion process. Han et al. [12] proposed a trusted multi-view classification framework that jointly optimizes view reliability and predictive confidence. Subsequent extensions explored dynamic fusion [13] and evolutionary optimization [21], enabling models to adaptively adjust view contributions in response to varying levels of trustworthiness and uncertainty. To further enhance uncertainty representation, [23] introduces a dynamic evidence decoupling method to separate view-specific reliability from task uncertainty, while [38] focuses on handling conflicting multi-view signals through reliability-aware contrastive learning. Other works have studied opinion aggregation [22] and application-specific settings [39].

While evidential learning has shown promise in classification tasks [1], its full potential in modeling complex emotional dynamics across modalities and time steps remains to be realized. Our work bridges this gap by introducing a structured decomposition of evidential uncertainty, capturing emotion-consistent, emotion-specific, and dynamic factors in a unified evidential framework.

3 ESED Framework

3.1 Task Formulation and Framework Overview

Given a conversation consisting of T utterances $\{U_1, U_2, \dots, U_n\}$, the goal of multimodal emotion recognition in conversation is to predict the emotion label $y_n \in \mathcal{Y}$ for each utterance U_n , where $\mathcal{Y}$ denotes a predefined emotion set. Each utterance U_n is associated with three modalities: text x_n^t , audio x_n^a , and visual x_n^v . The MERC task poses unique challenges due to emotional ambiguity, inter-modal conflicts, and the dynamic nature of emotional expression over time. Traditional multimodal models often rely on early or late fusion strategies, which inadequately capture uncertainty and overlook the differentiated roles of each modality under varying emotional conditions.

To address these issues, we propose ESED, a novel evidence-based framework that decomposes multimodal utterance-level representations into three interpretable components: emotion-consistent evidence, emotion-specific evidence, and dynamic evidence. The model first extracts robust unimodal features and transforms them into Dirichlet-based evidential representations to model uncertainty. These evidences are then projected into orthogonal subspaces to disentangle shared, modality-unique, and temporally evolving emotional cues. Each evidence component is guided by dedicated auxiliary objectives and dynamically reweighted based on its relevance to the emotional context. The final prediction is generated through an adaptive fusion of the decomposed evidence, enabling more robust and interpretable emotion recognition.

3.2 Evidential Uncertainty Modeling

To model the uncertainty inherent in multimodal emotion recognition, we adopt an evidential learning paradigm based on subjective logic. For each unimodal feature $x_n^m (m \in \{t, a, v\})$, we construct evidence $e_n^m \in \mathbb{R}_+^{|Y|}$ using a non-negative transformation, defined as $e_n^m = \text{ReLU}(f_\theta^m(x_n^m))$, where $f_\theta^m(\cdot)$ denotes a modality-specific neural encoder. The evidence vector e_n^m parameterizes a Dirichlet distribution $\text{Dir}(\alpha_n^m)$ over the class probabilities, with $\alpha_n^m = e_n^m + 1$. This formulation enables explicit modeling of uncertainty through the total evidence $S_n^m = \sum_{k=1}^{|Y|} \alpha_{n,k}^m$, where lower values of S_n^m correspond to higher epistemic uncertainty.

The expected predictive distribution is given by $\mathbb{E}[p_n^m] = \alpha_n^m / S_n^m$, and the associated uncertainty can be quantified using the vacuity $u_n^m = |Y| / S_n^m$. By representing each modality's belief as a Dirichlet distribution, our framework captures not only the confidence of class predictions but also the degree of uncertainty, allowing the model to defer or adapt when faced with ambiguous or conflicting signals. These evidential representations serve as the foundation for subsequent evidence decomposition, enabling interpretable and uncertainty-aware fusion across modalities.

3.3 Emotion-Specific Evidence Decomposition

To enhance interpretability and improve robustness under ambiguous or conflicting conditions, we decompose the unimodal evidential representation into semantically meaningful subspaces. Specifically, we project each modality's evidence into three disjoint components: emotion-consistent evidence, emotion-specific

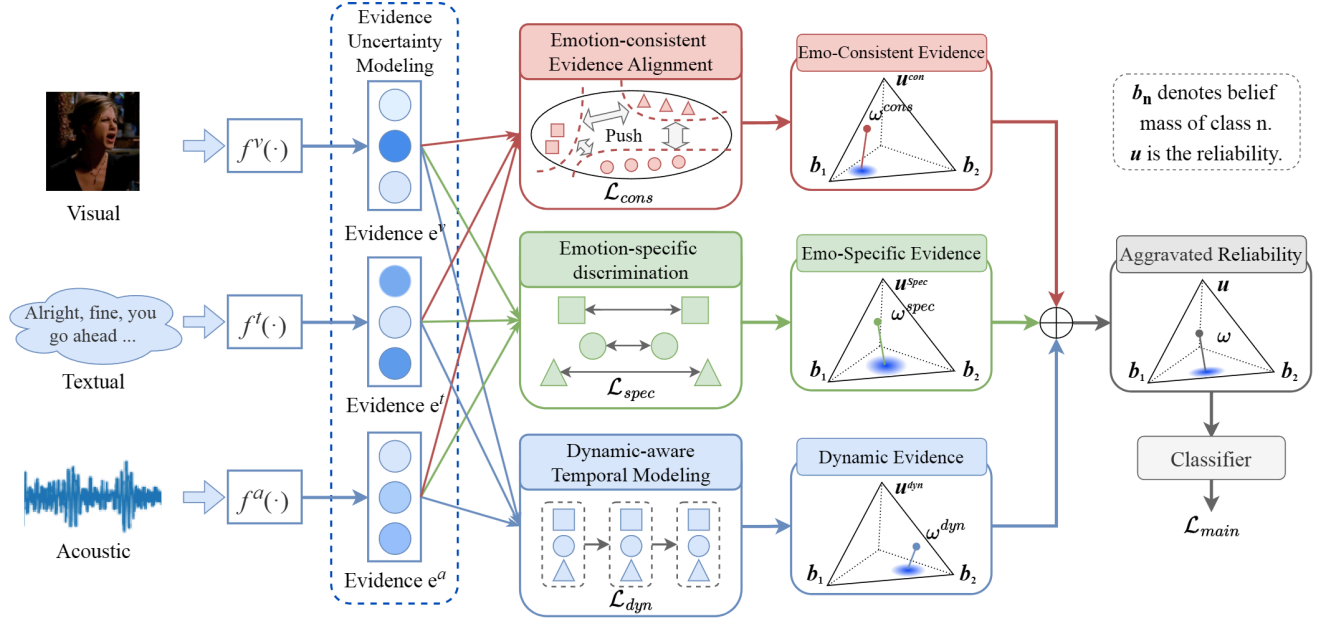


Figure 2: Overview of the proposed ESED framework. Given multimodal utterances, modality-specific encoders extract unimodal features, which are transformed into Dirichlet-based evidential representations. These are decomposed into three orthogonal subspaces: emotion-consistent, emotion-specific, and dynamic evidence.

evidence, and dynamic evidence, each corresponding to a distinct emotional cue structure.

3.3.1 Evidence Projection. Given the evidence vector $e_n^m \in \mathbb{R}_+^{|y|}$ for modality m at utterance U_n , we define three learnable linear projections $W_{cons}^m, W_{spec}^m, W_{dyn}^m \in \mathbb{R}^{|y| \times d}$ that map the original evidence to lower-dimensional latent spaces. The projected components are computed as:

$$z_{n,cons}^m = W_{cons}^m e_n^m, z_{n,spec}^m = W_{spec}^m e_n^m, z_{n,dyn}^m = W_{dyn}^m e_n^m. \quad (1)$$

These projections aim to capture (i) modality-invariant cues aligned across modalities, (ii) modality-specific expressions that characterize unique emotional signatures (e.g., pitch variation in audio or facial action in vision), and (iii) temporal fluctuations in emotion within conversations. The latent vectors $z_{n,\cdot}^m \in \mathbb{R}^d$ are later re-projected into the evidence space to reconstruct refined modality-aware evidence components for fusion.

3.3.2 Orthogonal Decoupling. To encourage semantic disentanglement and avoid redundant encoding across subspaces, we impose orthogonality constraints among the projected components. Specifically, we minimize the pairwise cosine similarity between every pair of projected vectors:

$$\mathcal{L}_{orth}^m = \sum_{(i,j) \in \mathcal{P}} \frac{\langle z_{n,i}^m, z_{n,j}^m \rangle}{\|z_{n,i}^m\|_2 \cdot \|z_{n,j}^m\|_2}, \quad (2)$$

where $z_{n,i}^m, z_{n,j}^m \in \mathbb{R}^d$ denote two projected components of modality m at time step n , $i, j \in \{cons, spec, dyn\}$ and $i \neq j$, $\langle \cdot, \cdot \rangle$ denotes

the inner product, and $\|\cdot\|_2$ is the Euclidean norm. Aggregated over all modalities and utterances, the full orthogonality loss is defined as:

$$\mathcal{L}_{orth} = \sum_{n=1}^T \sum_{m \in \{t,a,v\}} \mathcal{L}_{orth}^m. \quad (3)$$

By enforcing this structural constraint, this loss encourages each projected vector to occupy an independent subspace, effectively reducing semantic overlap and facilitating clearer attribution of modality roles.

3.4 Evidence-guided Auxiliary Supervision

To ensure that the decomposed evidence components effectively capture distinct emotional semantics, we introduce auxiliary supervision objectives that align each subspace with task-relevant signals. These objectives guide the model to learn modality-invariant consistency, modality-specific discriminability, and temporal emotional dynamics, which are crucial for robust multimodal emotion understanding.

3.4.1 Emotion-consistent Evidence Alignment. Emotion-consistent evidence aims to capture shared affective signals across modalities. To enforce semantic alignment, we minimize the pairwise distance between the emotion-consistent representations from different modalities at the same utterance step. Given emotion-consistent components $z_{n,cons}^t, z_{n,cons}^a, z_{n,cons}^v \in \mathbb{R}^d$, we define the alignment loss as:

$$\mathcal{L}_{align} = \sum_{(m_1, m_2) \in \mathcal{M}} \|z_{n,cons}^{m_1} - z_{n,cons}^{m_2}\|_2^2, \quad (4)$$

where $\mathcal{M} = \{(t, a), (t, v), (a, v)\}$. This objective enforces modality-invariant representations for shared emotional cues, reducing inter-modal conflicts and improving fusion robustness.

3.4.2 Emotion-specific discrimination. Emotion-specific evidence is expected to retain discriminative cues unique to each modality. To strengthen its emotion separability, we apply a classification head to each $z_{n,spec}^m$, encouraging it to independently predict the correct emotion label. The modality-specific discrimination loss is defined as:

$$\mathcal{L}_{spec} = \sum_{m \in \{t, a, v\}} \mathcal{L}_{CE}(f_{spec}^m(z_{n,spec}^m), y_n), \quad (5)$$

where $f_{spec}^m(\cdot)$ denotes the classifier applied to the specific subspace of modality m , and $\mathcal{L}_{CE}$ is the cross-entropy loss. This ensures that each modality-specific representation contributes unique and informative emotional cues.

3.4.3 Dynamic-aware Temporal Modeling. To model temporal variations in emotional dynamics, we pass the dynamic evidence components $\{Z_{1,dyn}^m, \dots, Z_{T,dyn}^m\}$ into a temporal encoder f_{dyn} , such as a transformer, and apply supervision on the aggregated output. The dynamic modeling loss is given by:

$$\mathcal{L}_{dyn} = \sum_{m \in \{t, a, v\}} \mathcal{L}_{CE}(f_{dyn}(\{z_{n,dyn}^m\}_{n=1}^T), \{y_n\}_{n=1}^T), \quad (6)$$

where the model captures temporal evolution across utterances, allowing dynamic patterns, such as emotional escalation or attenuation, to be properly encoded and predicted.

Together, these auxiliary objectives provide semantic guidance to disentangled evidence subspaces, reinforcing their intended roles and ensuring robust and interpretable multimodal emotion recognition in conversational settings.

3.5 Optimization Objective

To ensure that the final fused evidence effectively reflects the emotional semantics across modalities, we apply a primary classification loss on the integrated representation. Specifically, the three decomposed evidential components—emotion-consistent, emotion-specific, and dynamic evidence—are reprojected into the original evidence space and adaptively fused via an attention-based gating mechanism. The resulting fused evidence vector $e_n^{fuse} \in \mathbb{R}_+^{|y|}$ parameterizes a Dirichlet distribution, from which the expected class probability $\mathbb{E}[p_n^{fuse}]$ is computed as $\alpha_n^{fuse}/S_n^{fuse}$, where $\alpha_n^{fuse} = e_n^{fuse} + 1$ and $S_n^{fuse} = \sum_k \alpha_{n,k}^{fuse}$. The main objective is to minimize the cross-entropy between the expected predictive distribution and the ground-truth label y_n . Formally, the primary classification loss is defined as:

$$\mathcal{L}_{main} = \sum_{n=1}^T \mathcal{L}_{CE}(\mathbb{E}[p_n^{fuse}], y_n). \quad (7)$$

This loss term directly supervises the final prediction and ensures that the fused representation aligns with the true emotional state. By leveraging the evidential formulation, the loss also implicitly considers the uncertainty of the prediction, promoting cautious decision-making in ambiguous contexts.

Algorithm 1 ESED: Evidence-based Semantic Decomposition for Multimodal Emotion Recognition

Require: Utterance sequence $\{U_1, U_2, \dots, U_T\}$ with modalities x_n^m , $m \in \{t, a, v\}$

Ensure: Emotion predictions $\{\hat{y}_1, \dots, \hat{y}_T\}$

```

1: for  $n = 1$  to  $T$  do
2:   for all modality  $m \in \{t, a, v\}$  do
3:      $e_n^m \leftarrow \text{ReLU}(f_{\theta}^m(x_n^m))$   $\triangleright$  Unimodal evidence extraction
4:      $\alpha_n^m \leftarrow e_n^m + 1$   $\triangleright$  Dirichlet parameters
5:      $S_n^m \leftarrow \sum_k \alpha_{n,k}^m$ ,  $u_n^m \leftarrow |Y|/S_n^m$ 
6:      $p_n^m \leftarrow \alpha_n^m/S_n^m$   $\triangleright$  Expected class probability
7:      $Z_{n,cons}^m \leftarrow W_{cons}^m e_n^m$   $\triangleright$  Emotion-consistent evidence
8:      $Z_{n,spec}^m \leftarrow W_{spec}^m e_n^m$   $\triangleright$  Emotion-specific evidence
9:      $Z_{n,dyn}^m \leftarrow W_{dyn}^m e_n^m$   $\triangleright$  Dynamic evidence
10:   end for
11:    $\mathcal{L}_{orth}^n \leftarrow \text{OrthogonalLoss}(Z_{n,cons}^m, Z_{n,spec}^m, Z_{n,dyn}^m)$ 
12:    $\mathcal{L}_{align}^n \leftarrow \text{AlignLoss}(Z_{n,cons}^m, Z_{n,spec}^m)$ 
13:    $\mathcal{L}_{spec}^n \leftarrow \text{CE}(f_{spec}^m(Z_{n,spec}^m), y_n)$  for each  $m$ 
14:   end for
15:   for all modality  $m$  do
16:      $Z_{1:T,dyn}^m \leftarrow \{Z_{1,dyn}^m, \dots, Z_{T,dyn}^m\}$ 
17:      $\hat{y}_{1:T}^m \leftarrow f_{dyn}(Z_{1:T,dyn}^m)$ 
18:      $\mathcal{L}_{dyn}^m \leftarrow \text{CE}(\hat{y}_{1:T}^m, y_{1:T})$ 
19:   end for
20:    $\hat{y}_n \leftarrow \text{Fuse}(\{Z_{n,\cdot}^m\}_m)$   $\triangleright$  Adaptive fusion over all evidence
21:    $\mathcal{L}_{main} \leftarrow \text{CE}(\hat{y}_n, y_n)$ 
22:   Compute total loss:
23:    $\mathcal{L} = \mathcal{L}_{main} + \lambda_1 \mathcal{L}_{align} + \lambda_2 \mathcal{L}_{spec} + \lambda_3 \mathcal{L}_{dyn} + \lambda_4 \mathcal{L}_{orth}$ 
24:   Update model parameters to minimize  $\mathcal{L}$ 

```

In addition to the primary classification loss, the overall training objective incorporates multiple auxiliary supervision signals that reinforce the semantic disentanglement and task relevance of each evidence component. The total loss is defined as:

$$\mathcal{L} = \mathcal{L}_{main} + \lambda_1 \mathcal{L}_{align} + \lambda_2 \mathcal{L}_{spec} + \lambda_3 \mathcal{L}_{dyn} + \lambda_4 \mathcal{L}_{orth}, \quad (8)$$

Here, $\mathcal{L}_{align}$, $\mathcal{L}_{spec}$, and $\mathcal{L}_{dyn}$ are the auxiliary supervision losses introduced in Section 3.4, which guide the emotion-consistent, emotion-specific, and dynamic subspaces, respectively. $\mathcal{L}_{orth}$ is the orthogonality constraint loss introduced in Section 3.3 to encourage independence among evidence subspaces. The coefficients $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ are hyperparameters that balance the contribution of each term. This joint optimization encourages the model to learn disentangled, interpretable, and task-aware evidence representations, resulting in robust and semantically grounded multimodal emotion recognition.

4 Experiments

4.1 Datasets

We conduct experiments on two widely used multimodal emotion recognition datasets: **IEMOCAP** and **MELD**.

IEMOCAP consists of dyadic conversations totaling approximately 12 hours of audiovisual recordings, annotated with six

Table 1: Statistics of IEMOCAP and MELD datasets.

Dataset	Conversations			Utterances			Classes
	Train	Valid	Test	Train	Valid	Test	
IEMOCAP	120	31		5810	1623		6
MELD	1039	114	280	9989	1109	2610	7

emotion categories: *neutral, happy, sad, angry, excited, and frustrated*. Each utterance is accompanied by synchronized text, audio, and visual modalities. The dataset is particularly challenging due to its spontaneous, naturalistic dialogue, emotional ambiguity, and speaker variability.

MELD expands the setting to multiparty conversations, extracted from the TV series *Friends*. It includes seven emotion categories, with each utterance aligned across textual, audio, and visual modalities. The dataset introduces additional challenges such as speaker tracking, overlapping speech, and complex turn-taking behaviors in emotionally diverse interactions.

These datasets provide rigorous benchmarks for evaluating models in terms of multimodal integration, temporal emotion dynamics, and robustness to conversational complexity. Key statistics are summarized in Table 1.

4.2 Experimental Settings

All experiments are conducted on a server equipped with 2 NVIDIA RTX 3080 GPUs with CUDA 11.2, PyTorch 1.9.0, and Python 3.8.

Encoders. We adopt unimodal encoders to extract high-quality representations. Specifically, we use RoBERTa-base [24] for textual inputs, pre-trained WavLM-base [2] for acoustic features, and a ViT backbone [4] for visual modality. Each encoder is fine-tuned during training to adapt to the emotion recognition task. RoBERTa-base outputs a 768-dimensional vector per utterance, WavLM-base produces 1024-dimensional acoustic embeddings, and ViT yields 768-dimensional visual representations. These modality-specific embeddings are subsequently passed to our evidence disentanglement module for projection and fusion. All unimodal embeddings are projected to a unified 256-dimensional space before evidence decomposition.

Training Settings. The model is optimized using the Adam optimizer with an initial learning rate of 1×10^{-4} , accompanied by a weight decay of 1×10^{-5} to reduce overfitting. We apply a linear learning rate scheduler with 10% warm-up steps. The batch size is set to 32, and training is performed for a maximum of 50 epochs with early stopping based on validation loss (patience = 5).

Model Configuration. The dimensionality of all evidence projections is fixed at $d = 256$. Each modality-specific encoder f_{θ}^m consists of two fully connected layers with ReLU activations, producing modality-aware embeddings before projection. The projection matrices W_{cons}^m , W_{spec}^m , and W_{dyn}^m are implemented as learned linear layers, projecting the evidence vectors into mutually orthogonal semantic spaces of size d .

Temporal Modeling. To model the dynamics of emotional transitions across utterances, we apply a Transformer encoder with two layers and four attention heads. A dropout rate of 0.3 is employed within the Transformer to prevent overfitting.

Loss and Hyperparameters. The overall training objective includes multiple loss components with task-specific balancing coefficients λ_i , which are empirically tuned on the validation set.

4.3 Compared Methods

We compare our proposed method with several representative models in multimodal emotion recognition in conversations (MERC), covering both foundational architectures and recent state-of-the-art approaches:

- **DialogueRNN** [26]: An attentive recurrent neural network that models individual speaker states and contextual dependencies across conversational turns for emotion detection.
- **MMGCN** [16]: A multimodal graph convolutional network that builds modality-wise graphs over utterances to capture complex inter-modal and contextual relationships.
- **MMDFN** [15]: The Multimodal Dynamic Fusion Network dynamically integrates modality-specific and interaction-level features using a hierarchical fusion structure.
- **GA2MIF** [19]: A two-stage fusion framework that utilizes graph modeling and attention mechanisms to jointly leverage multimodal and multi-source information.
- **CMCF-SRNet** [40]: This model introduces cross-modality context fusion and semantic refinement to enhance the alignment of emotion-related cues across modalities.
- **DQ-Former** [17]: A querying transformer architecture that dynamically adjusts modality priorities and aligns representations with cognitive cues for robust emotion recognition.
- **AdaIGN** [33]: An adaptive graph learning model that captures evolving inter-modal dependencies through learnable graph structures tailored to conversation dynamics.

These baselines reflect the methodological progression in MERC, ranging from RNN-based temporal models to advanced graph-based and transformer-driven frameworks, providing a strong comparative foundation to evaluate our proposed evidential uncertainty decomposition method.

4.4 Main Results

We evaluate our proposed ESED method against strong baselines on the IEMOCAP and MELD datasets. Following standard practice, we report the weighted F1-score as the primary metric, which accounts for class imbalance and overall recognition performance. Additionally, we provide per-class F1-scores to analyze the fine-grained effectiveness of each model.

On the IEMOCAP dataset, ESED achieves the best weighted F1-score among all compared methods. Although it does not outperform all baselines on every individual class, it consistently ranks among the top across most emotion categories. Notably, it exhibits strong performance on high-variability emotions such as *Frustrated* and *Excited*, where many previous methods struggle. This suggests that ESED is particularly effective in modeling emotionally ambiguous or transitional expressions, which often involve subtle multimodal cues. The relatively high performance on *Neutral* and *Sad* also reflects its ability to maintain discriminative power across both high- and low-arousal emotions. While it does not lead in all classes, such as *Happy*, where DQ-Former shows an edge, its

Table 2: Per-class F1-score (%) and weighted F1-score on the IEMOCAP dataset.

Method	Neutral	Happy	Sad	Angry	Excited	Frustrated	Weighted F1
DialogueRNN [26]	57.89	32.20	80.26	62.82	73.87	59.76	62.89
MMGCN [16]	64.36	45.14	77.16	68.82	74.71	61.40	66.26
MMDFN [15]	66.42	42.22	78.98	69.77	75.56	66.33	68.18
GA2MIF [19]	68.38	46.15	84.50	70.29	75.99	66.49	70.00
CMCF-SRNet [40]	68.80	52.20	80.90	70.30	76.70	61.60	69.60
DQ-Former [17]	69.18	64.09	69.92	79.74	83.39	69.48	72.54
AdaIGN [33]	71.26	53.04	81.48	65.87	76.34	67.79	70.74
ESED (Ours)	73.02	57.62	83.19	76.25	84.27	70.15	72.66

Table 3: Per-class F1-score (%) and weighted F1-score on the MELD dataset.

Method	Neutral	Joy	Sad	Angry	Fear	Disgust	Surprise	Weighted F1
DialogueRNN [26]	76.79	50.92	20.41	45.52	-	-	47.69	57.66
MMGCN [16]	76.33	53.02	26.74	46.09	-	-	48.15	58.31
MMDFN [15]	77.76	54.78	22.94	47.82	-	-	50.69	59.46
GA2MIF [19]	76.92	51.87	27.18	48.52	-	-	49.08	58.94
CMCF-SRNet [40]	-	-	-	-	-	-	-	62.30
DQ-Former [17]	78.86	59.86	35.71	51.16	19.38	22.36	57.89	63.96
AdaIGN [33]	79.75	64.54	43.70	56.15	-	-	60.53	66.79
ESED (Ours)	81.04	66.70	41.25	59.83	26.49	28.72	62.47	68.21

balanced performance across the board contributes to the highest overall score.

In the MELD dataset, which features more diverse conversational contexts and a larger number of emotion categories, ESED again delivers the strongest weighted F1-score. It outperforms prior models in most classes, especially in previously underexplored or difficult categories such as Fear and Disgust, where many models do not report results or perform poorly. This reflects ESED’s capacity to generalize well in low-resource emotional classes, likely due to its fine-grained evidence decomposition and fusion mechanism. Moreover, even in dominant classes like Neutral and Joy, ESED retains a competitive or leading performance, indicating it does not compromise on frequent classes while boosting the rare ones.

The overall trends indicate that ESED’s strength lies not in peaking on a single class, but in maintaining consistently strong and stable performance across emotion types, regardless of their frequency or ambiguity. This suggests the model effectively balances expressive cues from different modalities and resolves potential conflicts between them, making it particularly suitable for real-world, multi-emotion, multimodal sentiment understanding tasks.

4.5 Ablation Study

4.5.1 Ablation Study on Evidence Subspaces. To examine the individual contribution of each evidence subspace in ESED, we perform an ablation study by selectively removing the emotion-consistent, emotion-specific, and dynamic subspaces. The results are reported on both IEMOCAP and MELD datasets in Table 4.

The most significant performance degradation occurs when the emotion-consistent subspace is removed, on both datasets. This

Table 4: Ablation study of evidence subspaces on IEMOCAP and MELD datasets (weighted F1-score %).

Model Variant	IEMOCAP	MELD
Full ESED	72.66	68.21
w/o Emotion-Consistent	69.84	65.92
w/o Emotion-Specific	71.37	67.10
w/o Dynamic	71.56	67.32

indicates that capturing shared semantic patterns across modalities is fundamental for robust multimodal emotion understanding. From a representational learning perspective, this component enforces intra-class compactness by aligning heterogeneous features into a modality-invariant latent space, thereby enhancing generalization and reducing noise from modality discrepancies.

Removing the emotion-specific subspace leads to a moderate but consistent drop in performance. This reveals the importance of preserving modality-unique cues, which often carry discriminative but complementary information. For example, acoustic features might encode prosodic variations that are not explicitly expressed in text, while facial expressions can signal subtle emotional states not easily captured by language. Disabling this component effectively reduces the model’s ability to leverage cross-modal redundancy and expressivity, both of which are essential in multimodal fusion.

The dynamic subspace, though having a smaller impact when removed, still contributes consistently to performance. This aligns with theoretical expectations in temporal and contextual modeling, where emotional meaning is influenced not only by isolated

utterances but also by their conversational context. By modeling evolving patterns across time, this component allows the system to adaptively attend to transient signals, improving the handling of emotion transitions and dialogue dynamics. Its contribution becomes particularly evident in dialogue-rich datasets like MELD.

Table 5: Impact of orthogonal constraint on model performance (weighted F1-score %).

Model Variant	IEMOCAP	MELD
With Orthogonal Loss	72.66	68.21
Without Orthogonal Loss	70.42	66.75

4.5.2 Ablation Study of Orthogonal Constraint. To evaluate the effectiveness of the orthogonal constraint, we compare model variants with and without this regularization term. The results in Table 5 consistently show a performance degradation across both datasets when the constraint is removed, indicating its importance in the model. This constraint helps separate the semantic spaces of different evidence types, reducing redundancy and encouraging each subspace to focus on distinct emotional patterns. Without it, the model may mix shared and specific cues, weakening its ability to extract complementary features from multimodal inputs. Thus, the orthogonal loss plays a key role in improving representation quality and overall recognition performance.

Table 6: Performance of ESED under different modality combinations (weighted F1-score %).

Modalities	IEMOCAP	MELD
Text (T) only	69.74	66.28
Audio (A) only	53.62	52.01
Visual (V) only	50.13	50.25
Text + Audio (T+A)	71.23	67.34
Text + Visual (T+V)	70.58	66.91
Audio + Visual (A+V)	55.77	53.69
Text + Audio + Visual (T+A+V)	72.66	68.21

4.5.3 Ablation Study of different modalities. To validate the effectiveness of multimodal fusion, we evaluate ESED under single-modality and combined-modality settings. Table 6 reports the quantitative results of the model using individual modalities—text, audio, and visual—as well as their pairwise and full fusion combinations.

Unimodal settings reveal that textual information alone achieves the strongest performance, indicating that language carries rich and explicit emotional semantics. In contrast, the audio and visual streams, while weaker in isolation, still contain valuable paralinguistic and behavioral cues such as tone and facial expressions. However, these nonverbal modalities are more sensitive to noise and contextual ambiguity, leading to limited discriminative power.

Bimodal combinations that include text (T+A or T+V) yield a noticeable performance boost compared to unimodal settings. This suggests that nonverbal modalities provide complementary signals

that enhance textual understanding. In contrast, the A+V combination offers only marginal gains over unimodal baselines, highlighting the asymmetric informativeness of different modalities.

The full trimodal configuration consistently achieves the highest performance, underscoring the synergistic effect of multimodal fusion. From a theoretical perspective, these results align with the principles of multiview learning and information integration theory. Each modality represents a distinct view of the emotional state; while views may vary in reliability, their combination, when properly weighted and aligned, yields a more holistic and robust latent representation. Furthermore, ESED’s dynamic and semantically-aware fusion mechanism reduces modality noise and mitigates cross-modal conflicts, thus preserving signal integrity.

4.6 Projection Dimension and Model Capacity

Table 7: F1 Performance of ESED under different projection dimensions and encoder sizes.

Projection Dim	Encoder Size	IEMOCAP	MELD
128	Small	67.13	63.57
256	Small	69.54	65.26
512	Small	71.07	65.80
128	Large	71.25	66.12
256	Large	72.66	68.21
512	Large	70.95	67.43

As shown in Table 7, we investigate how the projection dimension and encoder size jointly affect the expressiveness and generalization ability of ESED. Increasing the projection dimension yields steady improvements at low dimensionalities, indicating that a richer semantic evidential space allows the model to encode more fine-grained, modality-specific signals. However, this gain gradually saturates, and in some cases reverses, when the dimension becomes excessively high. This phenomenon reflects the classic bias-variance trade-off: overly large projection spaces increase the model’s capacity but may cause feature sparsity and overfitting under limited data, especially in emotionally ambiguous or low-resource settings like MELD. Larger encoder networks consistently outperform smaller ones, validating the need for sufficient nonlinear transformation power when modeling the semantic relationships between multimodal inputs. However, when the encoder is large and the projection space is also high-dimensional, performance either plateaus or slightly declines. This suggests a redundancy effect, where excessive model capacity leads to increased parameter coupling, optimization instability, and reduced ability to generalize.

4.7 Auxiliary Supervision Loss Weight Tuning

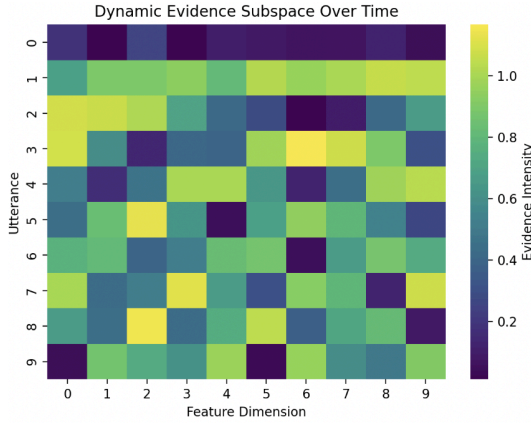
We evaluate the impact of varying weights for four auxiliary supervision signals targeting modality consistency, semantic disentanglement, and uncertainty control. Results in Table 8 show notable performance differences across configurations. When auxiliary losses are uniformly down-weighted, the model underutilizes crucial regularization, yielding weaker intermediate representations and suboptimal optimization dominated by the main task.

Table 8: F1 Performance of ESED under different auxiliary loss weight configurations.

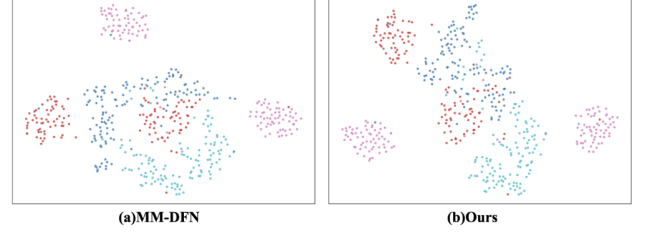
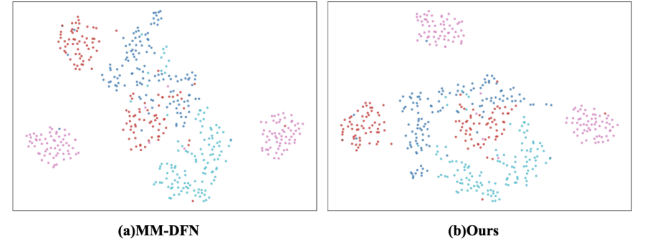
$(\lambda_1, \lambda_2, \lambda_3, \lambda_4)$	IEMOCAP	MELD	Remarks
(0.1,0.1,0.1,0.1)	68.48	64.29	Underweighted signals
(0.2,0.3,0.3,0.2)	72.66	68.21	Balanced supervision
(0.4,0.4,0.1,0.1)	69.14	65.76	Overfitting to partial tasks
(0.5,0.1,0.1,0.3)	69.73	66.23	Training instability

Conversely, excessive or imbalanced weights cause overfitting to certain tasks, disrupting synergy and inducing negative transfer, especially in multimodal settings with noisy or conflicting evidence. Optimal performance occurs with moderately balanced weights, enabling complementary guidance that improves semantic alignment, noise mitigation, and interpretability. This leads to more structured, generalizable representations across datasets.

4.8 Analysis of Dynamic Evidence Subspace Over Time

**Figure 3: Temporal heatmap of dynamic evidence intensity.**

To better understand how the model captures temporal variations in multimodal evidence, we visualize the dynamic evidence subspace across time steps, as shown in Figure 3. The heatmap presents the evidential activation values of different feature dimensions over time, where brighter regions correspond to higher activation intensities. From the figure, we observe that the model selectively activates different dimensions of the evidence space at different time steps, indicating its ability to dynamically attend to temporally relevant features. Notably, some dimensions maintain persistent activation, potentially reflecting core semantic features across the sequence, while others exhibit transient spikes, suggesting responsiveness to local cues or sudden emotional shifts. Such temporal variability highlights the model’s capacity to allocate evidential focus adaptively, which is crucial for capturing fine-grained emotional dynamics. This behavior aligns with human perception, where attention shifts based on contextual salience.

**Figure 4: t-SNE visualizations of learned feature embeddings on the MELD dataset. (a) MM-DFN. (b) Our method.****Figure 5: t-SNE visualizations of learned feature embeddings on the IEMOCAP dataset. (a) MM-DFN. (b) Our method.**

4.9 Qualitative Analysis

The t-SNE visualizations reveal substantial differences in embedding space structure between the proposed ESED method and the MM-DFN baseline. On MELD, the MM-DFN embeddings display dispersed clusters with significant overlap, suggesting that the learned representations fail to fully disentangle the emotional categories. This leads to ambiguity in classification and corresponds to the method’s lower accuracy. In contrast, ESED generates embeddings with tighter intra-class clustering and larger inter-class margins, demonstrating enhanced discriminative capacity. Similar patterns appear on the IEMOCAP dataset, where ESED’s embeddings manifest clearer boundaries between emotions. This improvement stems from the model’s incorporation of emotion-consistent and emotion-specific evidence subspaces, which effectively capture complementary emotion cues while reducing feature entanglement.

5 Conclusion

This work presents ESED, a novel framework designed to effectively leverage multi-evidence semantic disentanglement for multimodal emotion recognition in conversation. Through extensive experiments on MELD and IEMOCAP datasets, our method demonstrates superior performance compared to state-of-the-art methods. Despite these advancements, several limitations remain. The current approach relies on fixed weighting of auxiliary losses, which may not fully adapt to varying data distributions or modalities. Additionally, the model’s scalability and generalization to more diverse or large-scale datasets require further exploration.

Acknowledgments

This work was supported by the National Natural Science Foundation of China under Grant 52175493 and by the National Key Research and Development Program of China under Grant 2022YFB3206003.

GenAI Usage Disclosure

In this work, GenAI is applied in two ways. First, the Claude-3.5-Sonnet in Cursor is used to assist with code development, primarily focusing on the generation of visualization-related code. Second, Grok 3 and ChatGPT-4o are employed during the writing process to enhance the manuscript's spelling, grammar, punctuation, and clarity, functioning similarly to tools like Grammarly.

References

- [1] Feiyu Chen, Jie Shao, Anjie Zhu, Deqiang Ouyang, Xueliang Liu, and Heng Tao Shen. 2022. Modeling hierarchical uncertainty for multimodal emotion recognition in conversation. *IEEE Transactions on Cybernetics* 54, 1 (2022), 187–198.
- [2] Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, Xiong Xiao, et al. 2022. Wavlm: Large-scale self-supervised pre-training for full stack speech processing. *IEEE Journal of Selected Topics in Signal Processing* 16, 6 (2022), 1505–1518.
- [3] Vishal Chudasama, Purbayan Kar, Ashish Gudmalwar, Nirmesh Shah, Pankaj Wasnik, and Naoyuki Onoe. 2022. M2fnet: Multi-modal fusion network for emotion recognition in conversation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 4652–4661.
- [4] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xi-aohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. 2020. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929* (2020).
- [5] Soumya Dutta and Sriram Ganapathy. 2025. LLM supervised Pre-training for Multimodal Emotion Recognition in Conversations. *arXiv preprint arXiv:2501.11468* (2025).
- [6] Changzeng Fu, Fengkui Qian, Kaifeng Su, Yikai Su, Ze Wang, Jiaqi Shi, Zhigang Liu, Chaoran Liu, and Carlos Toshinori Ishi. 2025. HiMul-LGG: A hierarchical decision fusion-based local-global graph neural network for multimodal emotion recognition in conversation. *Neural Networks* 181 (2025), 106764.
- [7] Yumeng Fu, Junjie Wu, Zhongjie Wang, Meishan Zhang, Yulin Wu, and Bingquan Liu. 2025. BeMERC: Behavior-Aware MLLM-based Framework for Multimodal Emotion Recognition in Conversation. *arXiv preprint arXiv:2503.23990* (2025).
- [8] Yanping Fu, XiaoYuan Yan, Wei Chen, and Jun Zhang. 2025. Feature-Enhanced Multimodal Interaction model for emotion recognition in conversation. *Knowledge-Based Systems* 309 (2025), 112876.
- [9] Ankita Gandhi, Kinjal Adhvaray, Soujanya Poria, Erik Cambria, and Amir Hus-sain. 2023. Multimodal sentiment analysis: A systematic review of history, datasets, multimodal fusion methods, applications, challenges and future directions. *Information Fusion* 91 (2023), 424–444.
- [10] AV Geetha, T Mala, D Priyanka, and E Uma. 2024. Multimodal emotion recognition with deep learning: advancements, challenges, and future directions. *Information Fusion* 105 (2024), 102218.
- [11] Lili Guo, Yikang Song, and Shifei Ding. 2024. Speaker-aware cognitive network with cross-modal attention for multimodal emotion recognition in conversation. *Knowledge-Based Systems* 296 (2024), 111969.
- [12] Zongbo Han, Changqing Zhang, Huazhu Fu, and Joey Tianyi Zhou. 2021. Trusted Multi-View Classification. In *International Conference on Learning Representations*. <https://openreview.net/forum?id=OOSR8BzCn5>
- [13] Zongbo Han, Changqing Zhang, Huazhu Fu, and Joey Tianyi Zhou. 2022. Trusted multi-view classification with dynamic evidential fusion. *IEEE transactions on pattern analysis and machine intelligence* 45, 2 (2022), 2551–2566.
- [14] Samira Hazmoune and Fateh Bougamouza. 2024. Using transformers for multimodal emotion recognition: Taxonomies and state of the art review. *Engineering Applications of Artificial Intelligence* 133 (2024), 108339.
- [15] Dou Hu, Xiaolong Hou, Lingwei Wei, Lianxin Jiang, and Yang Mo. 2022. MM-DFN: Multimodal dynamic fusion network for emotion recognition in conversations. In *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 7037–7041.
- [16] Jingwen Hu, Yuchen Liu, Jiming Zhao, and Qin Jin. 2021. MMGCN: Multi-modal Fusion via Deep Graph Convolution Network for Emotion Recognition in Conversation. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*. 5666–5675.
- [17] Ye Jing and Xinpei Zhao. 2024. DQ-Former: Querying Transformer with Dynamic Modality Priority for Cognitive-aligned Multimodal Emotion Recognition in Conversation. In *Proceedings of the 32nd ACM International Conference on Multimedia*. 4795–4804.
- [18] Bobo Li, Hao Fei, Lizi Liao, Yu Zhao, Chong Teng, Tat-Seng Chua, Donghong Ji, and Fei Li. 2023. Revisiting disentanglement and fusion on modality and context in conversational multimodal emotion recognition. In *Proceedings of the 31st ACM International Conference on Multimedia*. 5923–5934.
- [19] Jiang Li, Xiaoping Wang, Guoqing Lv, and Zhigang Zeng. 2024. GA2MIF: Graph and Attention Based Two-Stage Multi-Source Information Fusion for Conversational Emotion Detection. *IEEE Transactions on Affective Computing* 15, 1 (2024), 130–143.
- [20] Qianer Li, Peijie Huang, Yuhong Xu, Jiawei Chen, Yuyang Deng, and Shangjian Yin. 2024. Generating and encouraging: An effective framework for solving class imbalance in multimodal emotion recognition conversation. *Engineering Applications of Artificial Intelligence* 133 (2024), 108523.
- [21] Xinyan Liang, Pinhan Fu, Yuhua Qian, Qian Guo, and Guoqing Liu. 2025. Trusted Multi-View Classification via Evolutionary Multi-View Fusion. In *The Thirteenth International Conference on Learning Representations*.
- [22] Wei Liu, Xiaodong Yue, Yufei Chen, and Thierry Denoeux. 2022. Trusted multi-view deep learning with opinion aggregation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 36. 7585–7593.
- [23] Ying Liu, Lihong Liu, Cai Xu, Xiangyu Song, Ziyu Guan, and Wei Zhao. 2024. Dynamic evidence decoupling for trusted multi-view learning. In *Proceedings of the 32nd ACM International Conference on Multimedia*. 7269–7277.
- [24] Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692* (2019).
- [25] Hui Ma, Jian Wang, Hongfei Lin, Bo Zhang, Yijia Zhang, and Bo Xu. 2023. A transformer-based model with self-distillation for multimodal emotion recognition in conversations. *IEEE Transactions on Multimedia* (2023).
- [26] Navonil Majumder, Soujanya Poria, Devamanyu Hazarika, Rada Mihalcea, Alexander Gelbukh, and Erik Cambria. 2019. Dialoguernn: An attentive rnn for emotion detection in conversations. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 33. 6818–6825.
- [27] Tao Meng, Yuntao Shou, Wei Ai, Nan Yin, and Keqin Li. 2024. Deep imbalanced learning for multimodal emotion recognition in conversations. *IEEE Transactions on Artificial Intelligence* (2024).
- [28] Tao Meng, Fuchen Zhang, Yuntao Shou, Wei Ai, Nan Yin, and Keqin Li. 2024. Revisiting multimodal emotion recognition in conversation from the perspective of graph spectrum. *arXiv preprint arXiv:2404.17862* (2024).
- [29] Tao Meng, Fuchen Zhang, Yuntao Shou, Hongen Shao, Wei Ai, and Keqin Li. 2024. Masked graph learning with recurrent alignment for multimodal emotion recognition in conversation. *IEEE/ACM Transactions on Audio, Speech, and Language Processing* (2024).
- [30] Murat Sensoy, Lance Kaplan, and Melih Kandemir. 2018. Evidential deep learning to quantify classification uncertainty. *Advances in neural information processing systems* 31 (2018).
- [31] Yuntao Shou, Tao Meng, Wei Ai, Nan Yin, and Keqin Li. 2023. A comprehensive survey on multi-modal conversational emotion recognition with deep learning. *arXiv preprint arXiv:2312.05735* (2023).
- [32] Yuntao Shou, Tao Meng, Wei Ai, Fuchen Zhang, Nan Yin, and Keqin Li. 2024. Adversarial alignment and graph fusion via information bottleneck for multimodal emotion recognition in conversations. *Information Fusion* 112 (2024), 102590.
- [33] Geng Tu, Tian Xie, Bin Liang, Hongpeng Wang, and Ruifeng Xu. 2024. Adaptive graph learning for multimodal conversational emotion detection. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 19089–19097.
- [34] Yusong Wang, Xuanye Fang, Hui Feng Yin, Dongyuan Li, Guoqi Li, Qi Xu, Yi Xu, Shuai Zhong, and Mingkun Xu. 2025. BIG-FUSION: Brain-Inspired Global-Local Context Fusion Framework for Multimodal Emotion Recognition in Conversations. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 1574–1582.
- [35] Yunxiao Wang, Meng Liu, Zhe Li, Yupeng Hu, Xin Luo, and Liqiang Nie. 2023. Unlocking the power of multimodal learning for emotion recognition in conversation. In *Proceedings of the 31st ACM International Conference on Multimedia*. 5947–5955.
- [36] Yuezhou Wu, Siling Zhang, and Pengfei Li. 2025. Multi-modal emotion recognition in conversation based on prompt learning with text-audio fusion features. *Scientific Reports* 15, 1 (2025), 8855.
- [37] Baijun Xie, Mariia Sidulova, and Chung Hyuk Park. 2021. Robust multimodal emotion recognition from conversation with transformer-based crossmodality fusion. *Sensors* 21, 14 (2021), 4913.
- [38] Cai Xu, Jiajun Si, Ziyu Guan, Wei Zhao, Yue Wu, and Xiyue Gao. 2024. Reliable conflictive multi-view learning. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 38. 16129–16137.
- [39] Cai Xu, Wei Zhao, Jinglong Zhao, Ziyu Guan, Xiangyu Song, and Jianxin Li. 2022. Uncertainty-aware multiview deep learning for internet of things applications. *IEEE Transactions on Industrial Informatics* 19, 2 (2022), 1456–1466.
- [40] Xiaoheng Zhang and Yang Li. 2023. A cross-modality context fusion and semantic refinement network for emotion recognition in conversation. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 13099–13110.